

Stable Chaos: Bounded Persistent Dynamics from Antagonistic and Imitative Coupling

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Abstract

Two coupling mechanisms pull a network of interacting units in opposite directions. Antagonistic (opposition) coupling, acting alone, drives a system into a rigid, frozen, low-entropy configuration. Imitative (likeness) coupling, acting alone, collapses the same system onto a single wandering consensus. We argue and demonstrate computationally that combining the two yields dynamics that are simultaneously bounded and persistently non-converging: trajectories never settle to a fixed point yet never diverge, and they occupy a large fraction of the accessible state space. We adopt the established term *stable chaos* of Politi and Torcini operationally to name this regime. Two models make the claim concrete. The Stable Chaos Model (SCM) is a discrete-time ring of two-channel nodes with a deterministic opposition rule and a stochastic imitation rule; ablating either mechanism removes it, as opposition-only freezes and imitation-only herds. The Tranception lattice is a frustrated Kuramoto lattice whose couplings are signed by a squared-distance orientation class, for which the 26-site neighborhood decomposes exactly as $26 = 6 + 12 + 8$. We give the model equations, the metrics used to quantify persistence and occupancy, an exact phasor superposition law that corrects a legacy transform-domain implementation, and figures from deterministic seed-42 runs. The scope is a model-level computational exploration; we make no continuum or hardware claims.

1 Introduction

Coupling between interacting units admits two elementary intuitions. *Antagonistic* coupling—repulsion, anti-alignment, or geometric frustration—tends to lock a system into a rigid, low-entropy arrangement, as in antiferromagnetic order. *Imitative* coupling—attraction, consensus, or phase alignment—tends to collapse a population onto a single state that then drifts together, as in herding. Taken separately, neither mechanism produces rich, sustained dynamics: one freezes and the other collapses. The thesis of this paper is that the two mechanisms, acting together, produce dynamics that are at once *bounded* and *persistently non-converging*. Opposition supplies the tension that prevents consensus; imitation supplies the mobility that prevents freezing; their competition keeps the system exploring a large portion of its bounded state space indefinitely.

We name this regime *stable chaos*, borrowing the term from Politi and Torcini [5], who use it for irregular dynamics that arise in systems that are linearly stable (non-positive maximal Lyapunov exponent) yet exhibit long-lived aperiodic behavior. Our usage is deliberately operational rather than a claim about Lyapunov spectra. We call a configuration operationally stable-chaotic when it satisfies two dynamical criteria on a compact state space: (i) *liveness*—the long-time mean step size stays bounded away from zero, so there is no fixed point or short limit cycle; and (ii) *diversity*—the units stay dispersed, with the cross-node variance held well above zero rather than collapsing onto

a single consensus. Boundedness is guaranteed by construction, and high state-space occupancy follows from liveness together with diversity. The distinction matters: a single herd can wander persistently and even visit a large fraction of the space over time, so occupancy alone does not separate stable chaos from consensus drift—only the conjunction of liveness and cross-node diversity does. These are model-level, computational criteria; we do not estimate Lyapunov exponents and make no continuum or hardware claims.

Two concrete models drive the paper. The Stable Chaos Model (SCM), Section 2, is a discrete-time ring of two-channel nodes formalizing a legacy prototype; it combines a deterministic opposition rule with a stochastic imitation rule, and its two switches let us ablate each mechanism independently. The Tranception lattice, Section 4, is a d -dimensional frustrated Kuramoto lattice whose couplings are signed by the geometric orientation class of each neighbor pair. Section 6 contributes an exact phasor superposition law used to correct a legacy transform-domain waveform implementation. Our contributions are: a symmetrized, reproducible formalization of the SCM with quantitative occupancy and persistence metrics (Sections 2–3); an orientation-class lattice with an exact neighborhood decomposition and its frustrated phase dynamics (Sections 4–5); and a closed-form waveform algebra (Section 6). All figures are produced by deterministic runs seeded at 42.

2 The Stable Chaos Model

2.1 State space and topology

The SCM is a ring of N nodes with N even and $N \geq 4$. Node $i \in \mathbb{Z}_N$ carries a state $\mathbf{s}_i = (A_i, B_i) \in [-1, 1]^2$; we call A and B its two channels. Three ring relations are used: the *opposite* $o(i) = (i + N/2) \bmod N$, the *left* neighbor $\ell(i) = (i + 1) \bmod N$, and the *right* neighbor $\rho(i) = (i - 1) \bmod N$. The sign function is the three-valued

$$\text{sgn}(x) = \begin{cases} -1, & x < 0, \\ 0, & x = 0, \\ +1, & x > 0, \end{cases} \quad (1)$$

and $\Pi(x) = \max(-1, \min(1, x))$ denotes clamping to $[-1, 1]$, applied componentwise to a state. The update step size is a fixed constant $\delta = 0.0987$ inherited from the legacy prototype. Initial states are drawn $A_i, B_i \sim \text{Uniform}(-1, 1)$ from a seeded generator, so every run is reproducible from its seed.

2.2 Update rule

The dynamics are synchronous: at tick t all nodes read the configuration $\{\mathbf{s}_j(t)\}$ and write $\{\mathbf{s}_j(t+1)\}$. Two increments act on each node, gated by boolean switches $\mathbb{K}_{\text{opp}}$ and $\mathbb{K}_{\text{imit}}$.

The *opposition* increment encodes the legacy repulsion semantics, in which the A -channel opposes the opposite node while the B -channel follows it:

$$\Delta_i^O(t) = \left(\delta \text{sgn}(A_i - A_{o(i)}), \text{sgn}(B_{o(i)} - B_i) \min(\delta, \tfrac{1}{2}|B_{o(i)} - B_i|) \right). \quad (2)$$

The B -channel follows to consensus, but its step is capped at half the gap $|B_{o(i)} - B_i|$. This cap is essential to the freeze result. A pure sign-step follow, $\delta \text{sgn}(B_{o(i)} - B_i)$, never reaches a fixed point: a mutually following pair overshoots the meeting point every tick and settles into a permanent $\pm\delta$ limit cycle about it, a discretization artifact rather than genuine motion. Capping the step at half the gap lets a diametric pair meet exactly at its midpoint and rest there, which is what

makes the opposition-only model *truly* freeze instead of jitter. The cap is inactive when the pair is far apart—it coincides with $\delta \operatorname{sgn}(B_{o(i)} - B_i)$ whenever $|B_{o(i)} - B_i| > 2\delta$. Acting alone this increment is deterministic: the A -channels amplify their differences until they saturate at ± 1 , while the B -channels are pulled together and rest, so the configuration relaxes onto a rigid, low-mobility arrangement.

The *imitation* increment is stochastic. At each node draw two independent variates $r_A, r_B \sim \text{Uniform}\{-1, 0, +1\}$ and set

$$\Delta_i^I(t) = \begin{cases} \delta(-1, -1), & (r_A, r_B) = (+1, +1), \\ \delta(+1, +1), & (r_A, r_B) = (-1, -1), \\ \delta r_A (\operatorname{sgn}(A_{\ell(i)} - A_i), \operatorname{sgn}(B_{\ell(i)} - B_i)), & r_A \neq 0 \text{ (else)}, \\ \delta r_B (\operatorname{sgn}(A_{\rho(i)} - A_i), \operatorname{sgn}(B_{\rho(i)} - B_i)), & r_A = 0, r_B \neq 0, \\ (0, 0), & r_A = r_B = 0. \end{cases} \quad (3)$$

The first two cases are joint retreat and joint advance along the state diagonal. Otherwise the node moves toward $(+1)$ or away from (-1) its left neighbor when $r_A \neq 0$, or its right neighbor when only $r_B \neq 0$; the coefficient r_A (resp. r_B) flips “toward” into “away.” Acting alone this is a stochastic consensus drift: nodes chase and flee their ring neighbors, wandering without the tension needed to keep exploring.

The full synchronous update is *sequential*: the opposition increment is applied first to form a clamped working self $\mathbf{w}_i$, and the imitation increment then acts on that working self, while every *neighbor* state is still read from the tick- t configuration:

$$\begin{aligned} \mathbf{w}_i &= \Pi \left[\mathbf{s}_i(t) + \mathbb{K}_{\text{opp}} \Delta_i^O(\mathbf{s}(t)) \right], \\ \mathbf{s}_i(t+1) &= \Pi \left[\mathbf{w}_i + \mathbb{K}_{\text{imit}} \Delta_i^I(\mathbf{w}_i, \mathbf{s}(t)) \right]. \end{aligned} \quad (4)$$

Here Δ_i^O is evaluated entirely at $\mathbf{s}(t)$, whereas in Δ_i^I the node’s own references A_i, B_i of Equation (3) are taken at the working self $\mathbf{w}_i$ while the neighbor references $A_{\ell(i)}, B_{\ell(i)}, A_{\rho(i)}, B_{\rho(i)}$ remain at tick t ; imitation’s toward/away sign is therefore measured against the post-opposition working self, not against $\mathbf{s}_i(t)$. The clamp Π is applied at each sub-step, so an intermediate excursion past $[-1, 1]$ is never carried into the next term. Because both increments read all neighbors synchronously at tick t and are applied to every node in the same fixed order, the update is independent of node iteration order. The legacy four-node prototype applied heterogeneous, iteration-order-dependent rules; Equations (2)–(4) are their symmetrized, order-independent formalization (the per-node quirks are documented separately in the repository legacy notes).

2.3 Ablations

Equation (4) defines three variants through the switches. The *full* model runs both increments ($\mathbb{K}_{\text{opp}} = \mathbb{K}_{\text{imit}} = 1$). The *opposition-only* model keeps only Δ^O and is deterministic. The *imitation-only* model keeps only Δ^I and is purely stochastic. These three variants are the empirical basis for the thesis: only the full model is expected to remain both persistently mobile and dispersed across $[-1, 1]^2$, whereas opposition-only freezes and imitation-only herds into a single cluster.

2.4 Metrics

A trajectory is stored as an array of shape $(T, N, 2)$. We quantify it with the following scalars, defined here exactly as they are computed. For a configuration at tick t write $\chi_i(t)$ for channel

$\chi \in \{A, B\}$ of node i and $\bar{\chi}(t) = \frac{1}{N} \sum_i \chi_i(t)$.

The per-tick *node variance*, averaged over channels, measures cross-node dispersion:

$$V(t) = \frac{1}{2} \sum_{\chi \in \{A, B\}} \frac{1}{N} \sum_i (\chi_i(t) - \bar{\chi}(t))^2. \quad (5)$$

The *mean step size*, the mean Euclidean per-node displacement over the final w transitions, measures persistence; values near zero indicate a frozen state:

$$S_w = \frac{1}{wN} \sum_{t=T-w}^{T-1} \sum_i \|\mathbf{s}_i(t+1) - \mathbf{s}_i(t)\|_2. \quad (6)$$

The *occupancy entropy* bins the pooled (A, B) samples over all nodes and ticks into a $b \times b$ grid on $[-1, 1]^2$, forms the empirical distribution $p_k = n_k / \sum_k n_k$, and normalizes the Shannon entropy (in nats) by $\ln b^2$ so that $\hat{H} \in [0, 1]$:

$$\hat{H} = \frac{-\sum_{k: p_k > 0} p_k \ln p_k}{\ln b^2}. \quad (7)$$

For a one-dimensional series x_t the normalized *autocorrelation* at lag k is

$$\rho(k) = \frac{\sum_t (x_t - \bar{x})(x_{t+k} - \bar{x})}{\sum_t (x_t - \bar{x})^2}, \quad \rho(0) = 1. \quad (8)$$

Two aggregates carry over to the phase models. The *Kuramoto order parameter* of N phases φ_j is

$$r = \left| \frac{1}{N} \sum_j e^{i\varphi_j} \right| \in [0, 1], \quad (9)$$

and the *acumen*, the legacy global aggregate, is the mean state $\bar{\mathbf{s}} = \frac{1}{N} \sum_i \mathbf{s}_i \in \mathbb{R}^2$.

3 SCM Results

All SCM figures use $N = 4$, $T = 2000$ ticks, and seed 42. Figure 1 plots the two channels of every node over time. The full model produces channels that keep moving throughout the run without collapsing to a constant or to a short cycle, the visual signature of persistence. Figure 2 shows the same run as (A, B) paths per node with start and end markers; the nodes migrate to the two vertical edges $A = \pm 1$ and then keep sweeping in B along them, rather than parking at a single corner or converging to a common point.

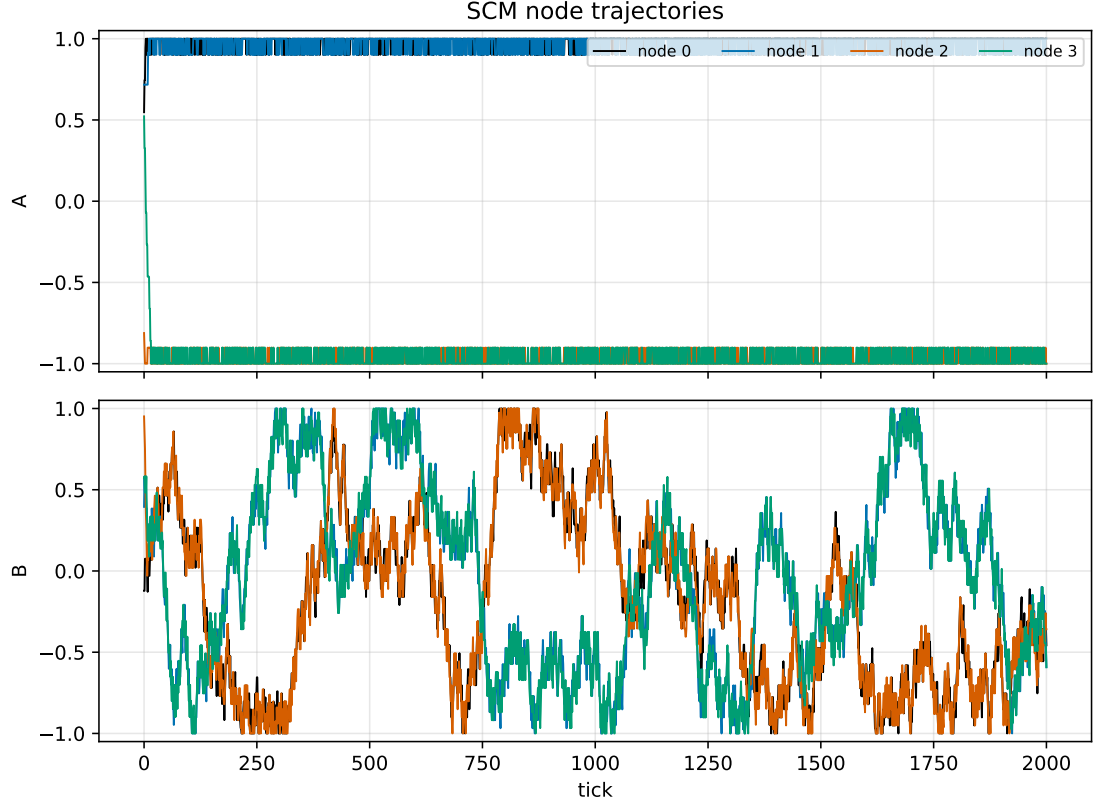


Figure 1: SCM channel trajectories $A_i(t)$ (top) and $B_i(t)$ (bottom) for the full model, $N = 4$, seed 42. Channels remain active for the full run rather than settling.

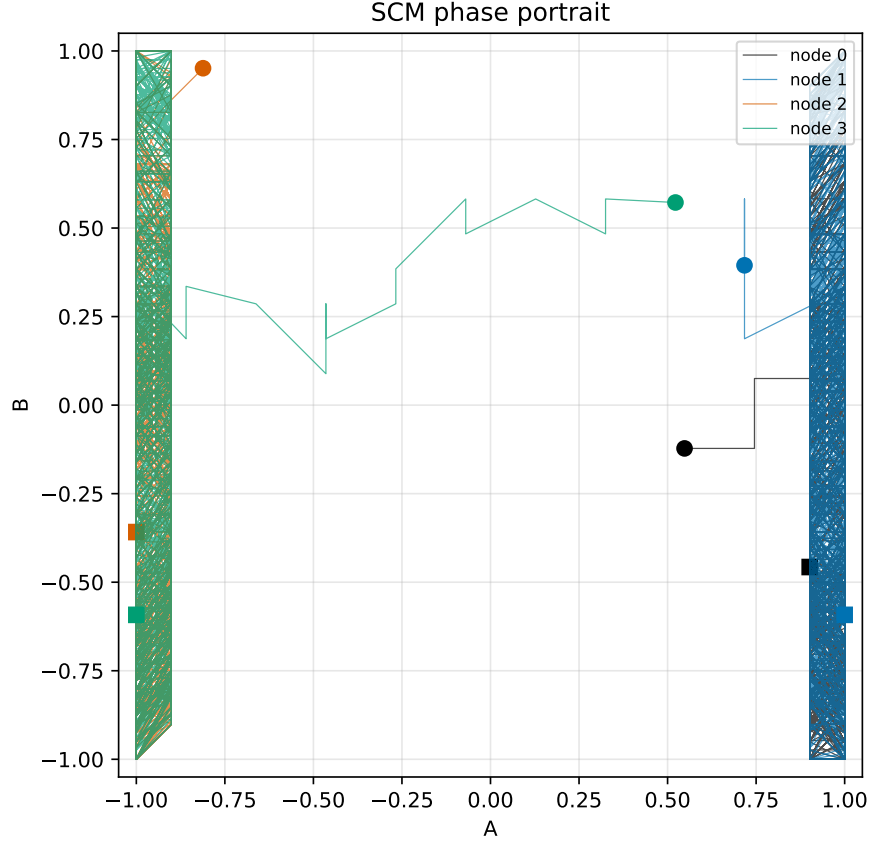


Figure 2: Phase portrait: per-node paths in the (A, B) plane with start (open) and end (filled) markers. The nodes settle onto the two vertical bands $A = \pm 1$ while continuing to wander in B , rather than converging to a single fixed point.

The ablation, Figure 3, contrasts the three variants directly. The opposition-only model relaxes to a rigid configuration: its late-time mean step size falls to zero and it freezes, yet its nodes remain dispersed—the antagonistic A -channels saturate against opposite clamps, so the cross-node variance stays high while nothing moves. The imitation-only model is the mirror image: it stays mobile but herds, so its cross-node variance collapses as the nodes chase one another into a common wandering cluster. The full model is the only variant that is simultaneously live and dispersed. Figure 4 shows the pooled two-dimensional occupancy of the full model: two high-count bands at $A = \pm 1$ with the B -channel sweeping the full range $[-1, 1]$ inside them—frozen antagonistic order in A coexisting with persistent wandering in B .

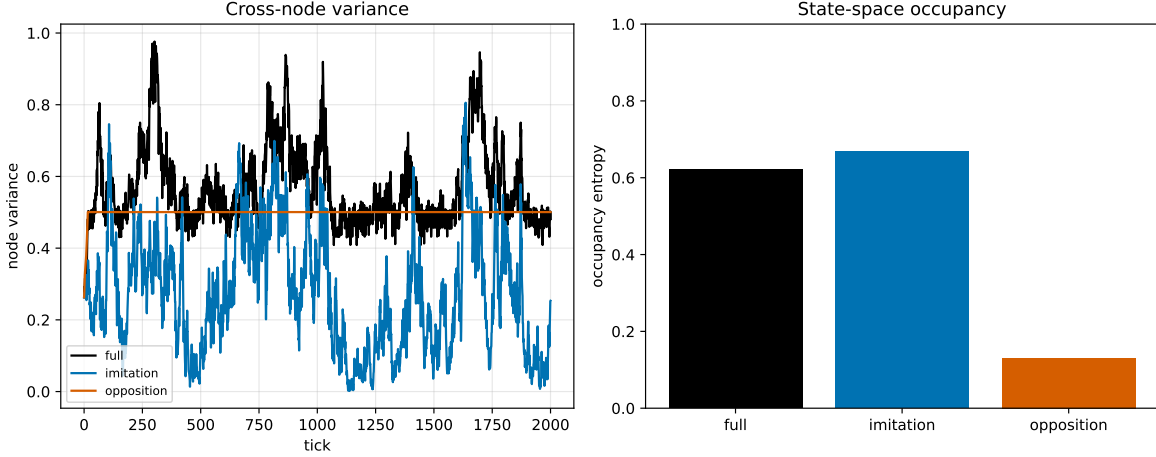


Figure 3: Ablation. Left: node variance $V(t)$ for the full, opposition-only, and imitation-only variants. Right: normalized occupancy entropy $\hat{H}$ per variant. Opposition-only freezes (variance stays high but nothing moves); imitation-only in fact reaches the *highest* occupancy entropy, but as a consensus herd whose cross-node variance collapses (left panel); the full model pairs comparable entropy with sustained cross-node diversity.

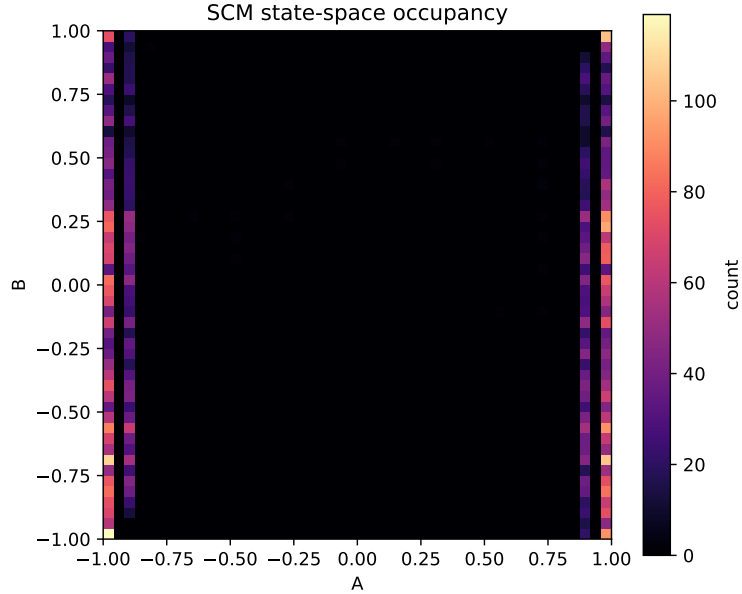


Figure 4: Pooled occupancy histogram of the full model over all nodes and ticks. Occupancy concentrates in two vertical bands at $A = \pm 1$ —the antagonistic A -channel pins to the clamps—while the B -channel sweeps the full range $[-1, 1]$ within them. The high normalized entropy of Equation (7) ($\hat{H} = 0.622$) comes from this B -spread inside the two bands, not from broad interior coverage: frozen order in A coexisting with persistent wandering in B .

Table 1 reports the three summary scalars for each variant from the seed-42 runs, and they tell a more careful story than occupancy alone would. Occupancy entropy separates only the frozen

opposition-only variant ($\hat{H} = 0.131$) from the two live variants; between those two it does *not* favor the full model. The imitation-only herd in fact wanders through slightly *more* of the state space over time ($\hat{H} = 0.668$ versus 0.622 for the full model), because, being unanchored, its consensus cluster drifts freely across the square. What distinguishes the variants is the *conjunction* of two criteria. The first is liveness, the late-time mean step size: it is 0.113 for the full model and 0.120 for imitation-only, but 0.000 for opposition-only, which is frozen. The second is diversity, the cross-node variance over the last 500 ticks: it is 0.579 for the full model but collapses to 0.261 for the imitation-only herd, while opposition-only stays dispersed at 0.501 because its A -channels are pinned at opposite clamps. Opposition-only thus fails liveness, imitation-only fails diversity, and only the full model satisfies both—persistent motion together with sustained cross-node spread. We restrict quantitative claims to Table 1 and to this two-criteria separation; all other claims are the qualitative ones the figures make directly.

Table 1: SCM summary metrics, seed 42, $N = 4$, $T = 2000$. Mean step size S_w over the final $w = 200$ transitions; occupancy entropy $\hat{H}$ with $b = 16$ bins, normalized to $[0, 1]$; cross-node variance $\bar{V}$ time-averaged over the final 500 ticks.

Variant	Mean step size S_{200}	Occupancy entropy $\hat{H}$	Cross-node variance $\bar{V}$ (last 500)
Full (opposition + imitation)	0.113	0.622	0.579
Opposition only	0.000	0.131	0.501
Imitation only	0.120	0.668	0.261

4 The Tranception Lattice

4.1 Orientation classes

The second model places phase oscillators on a d -dimensional cubic lattice, $d \in \{1, 2, 3\}$, of side L , with L^d nodes indexed $0, \dots, L^d - 1$ and integer coordinates recovered by the usual mixed-radix map. Each node is embedded in $\mathbb{Z}^3$ by padding absent axes with zeros so that a single classifier serves every dimension. For a coordinate offset $\mathbf{c} = (c_x, c_y, c_z) \in \{-1, 0, +1\}^3$ we classify the neighbor pair by squared Euclidean distance $\|\mathbf{c}\|^2 = c_x^2 + c_y^2 + c_z^2$:

$$\text{class}(\mathbf{c}) = \begin{cases} \text{SELF}, & \|\mathbf{c}\|^2 = 0, \\ \text{ORTHOGONAL}, & \|\mathbf{c}\|^2 = 1, \\ \text{ADJACENT}, & \|\mathbf{c}\|^2 = 2, \\ \text{POLAR}, & \|\mathbf{c}\|^2 = 3, \end{cases} \quad (10)$$

and the classifier is undefined (no coupling) whenever any $|c_k| > 1$ or $\|\mathbf{c}\|^2 > 3$, i.e. outside the 27-site cube. Couplings are stored as unique undirected pairs with the SELF class excluded; on a torus (the *toroidal* option) coordinates wrap modulo L . The following exact decomposition motivates the naming.

Proposition 1 (Neighborhood decomposition). *The 26 non-central sites of the cube $\{-1, 0, +1\}^3$ split by orientation class as*

$$26 = \underbrace{6}_{\text{ORTHOGONAL}} + \underbrace{12}_{\text{ADJACENT}} + \underbrace{8}_{\text{POLAR}}. \quad (11)$$

Proof. Because each $c_k \in \{-1, 0, +1\}$ we have $c_k^2 \in \{0, 1\}$, so $\|\mathbf{c}\|^2$ equals the number m of nonzero coordinates. Sites at squared distance m are those with exactly m nonzero coordinates, each chosen as ± 1 : there are $\binom{3}{m}2^m$ of them. Thus $m = 1$ gives $\binom{3}{1}2 = 6$, $m = 2$ gives $\binom{3}{2}2^2 = 12$, and $m = 3$ gives $\binom{3}{3}2^3 = 8$, summing to 26. $\square$

An interior node of a three-dimensional lattice therefore has exactly 6 orthogonal, 12 adjacent, and 8 polar neighbors; Figure 5 renders the three classes. Corner and edge nodes on a non-toroidal lattice have fewer, while on a torus every node recovers the full 6/12/8 split.

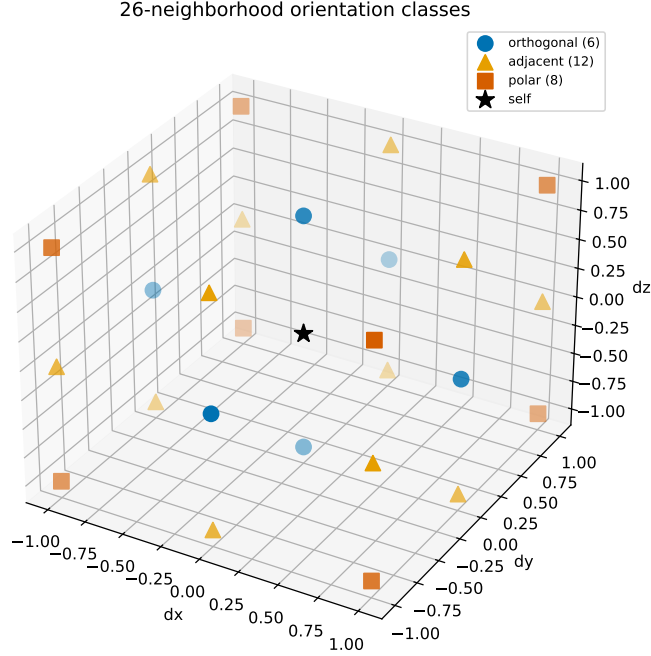


Figure 5: The 26-site neighborhood of Proposition 1, colored by orientation class: 6 orthogonal (face), 12 adjacent (edge), and 8 polar (corner) neighbors of the central node.

4.2 Signed phase dynamics

Each node i carries a phase $\varphi_i \in [0, 2\pi)$ initialized uniformly, and a natural frequency $\omega_i \sim \mathcal{N}(\mu, \sigma^2)$. Each orientation class c is assigned a coupling constant K_c , with the polar class taken negative to introduce frustration. Writing $N_c(i)$ for the class- c neighbors of i , the mean-field-normalized dynamics are the frustrated Kuramoto lattice

$$\dot{\varphi}_i = \omega_i + \sum_c \frac{K_c}{|N_c(i)|} \sum_{j \in N_c(i)} \sin(\varphi_j - \varphi_i), \quad (12)$$

where classes with $|N_c(i)| = 0$ contribute no term. The per-class normalization makes the drive comparable across nodes of different degree. We integrate Equation (12) with a forward Euler step of size Δt and wrap onto the circle,

$$\varphi_i(t + \Delta t) = [\varphi_i(t) + \Delta t \dot{\varphi}_i(t)] \bmod 2\pi, \quad (13)$$

and read out global coherence with the Kuramoto order parameter r of Equation (9). With $K_{\text{orth}} > 0$ and $K_{\text{adj}} > 0$ the orthogonal and adjacent classes favor alignment; the negative polar coupling $K_{\text{polar}} < 0$ opposes it, and the balance between the two is what the next section sweeps.

5 Lattice Results

We integrate a $6 \times 6 \times 6$ toroidal lattice ($d = 3$) for 4000 steps at $\Delta t = 0.05$ with $K_{\text{orth}} = 1$, $K_{\text{adj}} = 0.5$, and varying polar coupling, all seeded at 42; the experiments use a genuinely three-dimensional lattice because the polar orientation class is empty in lower dimensions, as the Remark below makes precise. Figure 6 tracks the order parameter $r(t)$ for $K_{\text{polar}} \in \{0, -0.5, -1\}$. Without frustration ($K_{\text{polar}} = 0$) the alignment terms dominate and r climbs to near-total coherence, $r \approx 0.996$: the lattice synchronizes. Switching on the polar coupling collapses coherence: at $K_{\text{polar}} = -1$ the frustration overwhelms alignment and the time-averaged order parameter falls to $r \approx 0.004$, an essentially incoherent field. The intermediate coupling $K_{\text{polar}} = -0.5$ still synchronizes, only more slowly, plateauing near $r \approx 0.98$; the transition to incoherence sits between -0.5 and -0.75 in the sweep of Figure 7.

Remark (Dimensional threshold). Polar frustration is intrinsically three-dimensional in this framework. A polar neighbor is a site at squared distance three, which requires three simultaneously nonzero coordinates; in one or two dimensions no such offset exists, so the polar class is empty and K_{polar} multiplies an absent term. The frustration is therefore *vacuous* below three dimensions: on a lattice with $d \leq 2$ the polar coupling has no effect whatsoever. We confirmed this with a two-dimensional control, a 12×12 torus under otherwise identical settings, whose time-averaged order parameter is identical to four decimals with and without polar coupling ($r = 0.9905$ for both $K_{\text{polar}} = 0$ and $K_{\text{polar}} = -1$). This is why the sweep and order experiments use a three-dimensional lattice, where the eight polar corners are present and can frustrate the six orthogonal and twelve adjacent alignment bonds.

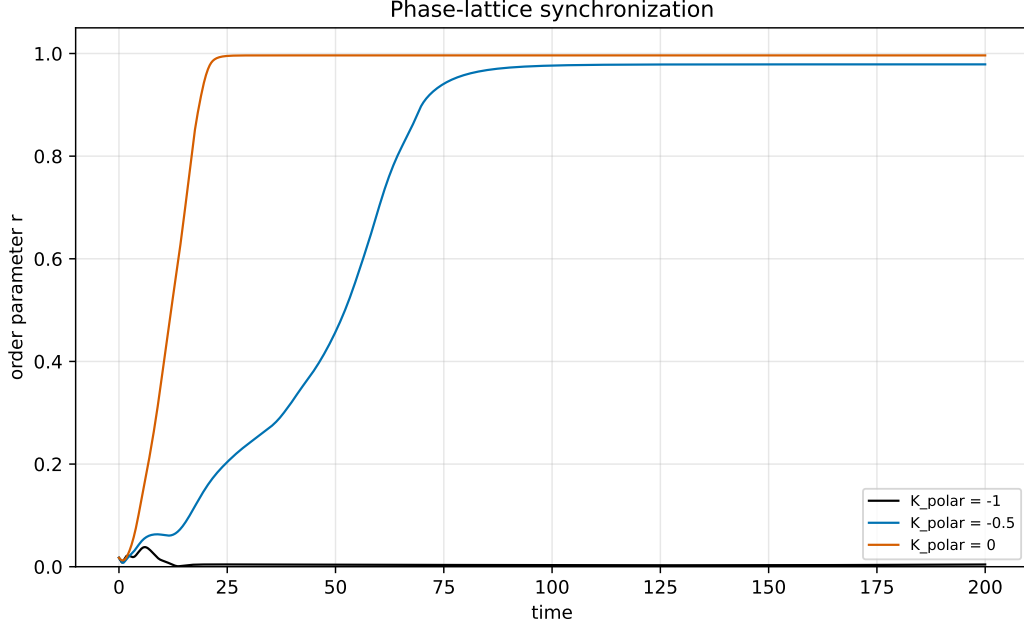


Figure 6: Order parameter $r(t)$ on a $6 \times 6 \times 6$ torus for three polar couplings. Unfrustrated coupling ($K_{\text{polar}} = 0$) synchronizes ($r \approx 0.996$); strong frustration ($K_{\text{polar}} = -1$) collapses coherence ($r \approx 0.004$).

Figure 7 sweeps K_{polar} across $\text{linspace}(-2, 0.5, 11)$ and reports the time-averaged $\langle r \rangle$ over the last quarter of each run. Coherence is high for non-negative polar coupling and is sharply suppressed as K_{polar} becomes more negative, tracing the transition from synchronization to a frustrated, incoherent regime. Figure 8 contrasts the two regimes directly through the middle z -slice of the $6 \times 6 \times 6$ torus after 4000 steps: the synchronized field ($K_{\text{polar}} = 0$, left) is nearly a single hue, whereas the frustrated field ($K_{\text{polar}} = -1$, right) is a disordered patchwork of phases, the spatial signature of the collapse in coherence.

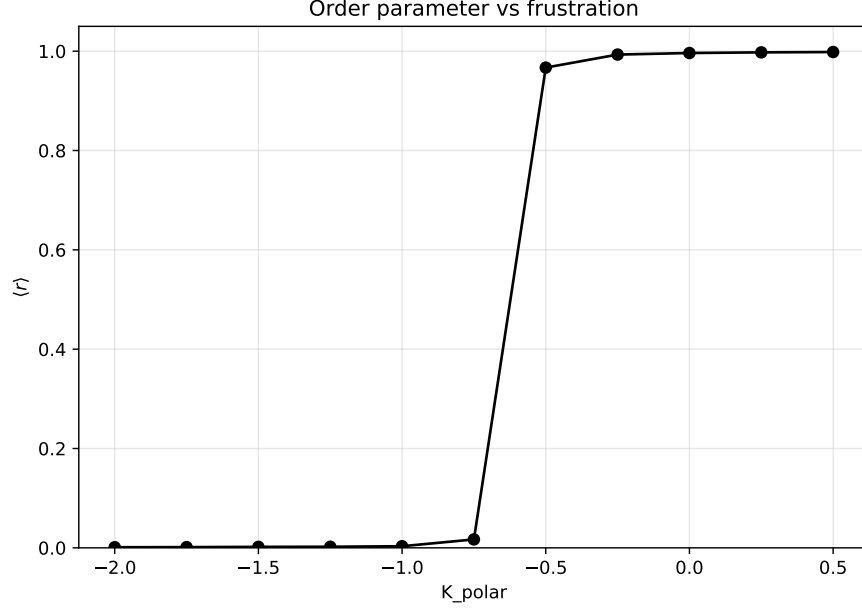


Figure 7: Time-averaged order parameter $\langle r \rangle$ versus polar coupling K_{polar} on the $6 \times 6 \times 6$ torus. Synchronization is suppressed as frustration increases.

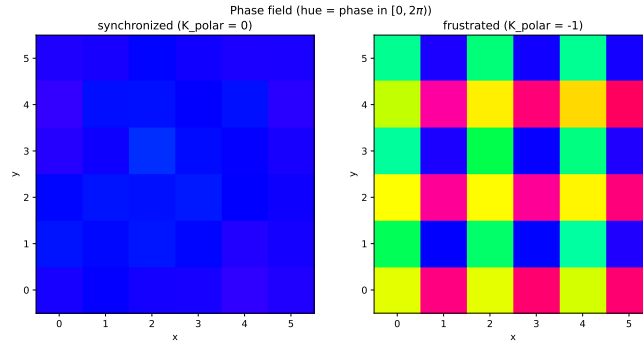


Figure 8: Middle z -slice of the $6 \times 6 \times 6$ toroidal phase field after 4000 steps, colored by phase. Left: synchronized regime ($K_{\text{polar}} = 0$), a near-uniform hue. Right: frustrated regime ($K_{\text{polar}} = -1$), a disordered patchwork consistent with $r \approx 0.004$.

The two models thus tell the same story with different vocabularies. In the SCM the deterministic opposition rule freezes and the stochastic imitation rule herds, while their combination stays both mobile and diverse. In the lattice, alignment couplings synchronize and the negative polar coupling frustrates, while their combination holds the field in a partially ordered, non-uniform state. Antagonism and imitation are again in tension, and the tension is what sustains the dynamics.

6 Waveform Algebra

The legacy Tranception code combined waveforms by transforming sampled buffers to the frequency domain, adding, and transforming back. That round trip is exact only up to sampling and truncation

error and leaks energy across frequency bins. For sinusoids of a common frequency the correct combination is closed-form and exact, given by phasor addition.

Lemma 1 (Phasor superposition). *For two sinusoids of common frequency f , $y_k(t) = a_k \sin(2\pi ft + \phi_k)$ with $a_k \geq 0$, the sum is a single sinusoid of the same frequency,*

$$y_1(t) + y_2(t) = a_3 \sin(2\pi ft + \phi_3), \quad (14)$$

whose amplitude and phase are those of the complex phasor sum $a_1 e^{i\phi_1} + a_2 e^{i\phi_2} = a_3 e^{i\phi_3}$:

$$a_3 = \sqrt{a_1^2 + a_2^2 + 2a_1 a_2 \cos(\phi_1 - \phi_2)}, \quad (15)$$

$$\phi_3 = \text{atan2}(a_1 \sin \phi_1 + a_2 \sin \phi_2, a_1 \cos \phi_1 + a_2 \cos \phi_2). \quad (16)$$

Subtraction replaces $a_2 e^{i\phi_2}$ by $-a_2 e^{i\phi_2}$, equivalently $\phi_2 \mapsto \phi_2 + \pi$.

Proof. Write $\theta = 2\pi ft$ and use $\sin \alpha = \text{Im } e^{i\alpha}$. Then $y_1 + y_2 = \text{Im}(e^{i\theta}(a_1 e^{i\phi_1} + a_2 e^{i\phi_2})) = \text{Im}(e^{i\theta} a_3 e^{i\phi_3}) = a_3 \sin(\theta + \phi_3)$. The modulus and argument of the phasor sum give a_3 and ϕ_3 ; the modulus expands to the stated law by the cosine rule. $\square$

The implementation follows the lemma. Same-frequency addition and subtraction use the phasor law exactly, and a frequency mismatch is rejected with an explicit error rather than silently approximated. Multiplication keeps the legacy convention—product of amplitudes and sum of phases—which is a design choice rather than the trigonometric product-to-sum identity, and it is documented as such. The resonant frequency of two waveforms is f when they agree and the geometric mean $\sqrt{f_1 f_2}$ otherwise; the reception level is $a_1 a_2$ at equal frequency and zero otherwise, preserving legacy semantics. Waveforms of arbitrary, possibly distinct frequencies are combined by **superpose**, which sums samples directly in the time domain; the resulting amplitude never exceeds the sum of the parts. Figure 9 shows two components and their superposition, including the beat envelope that arises when their frequencies are close.

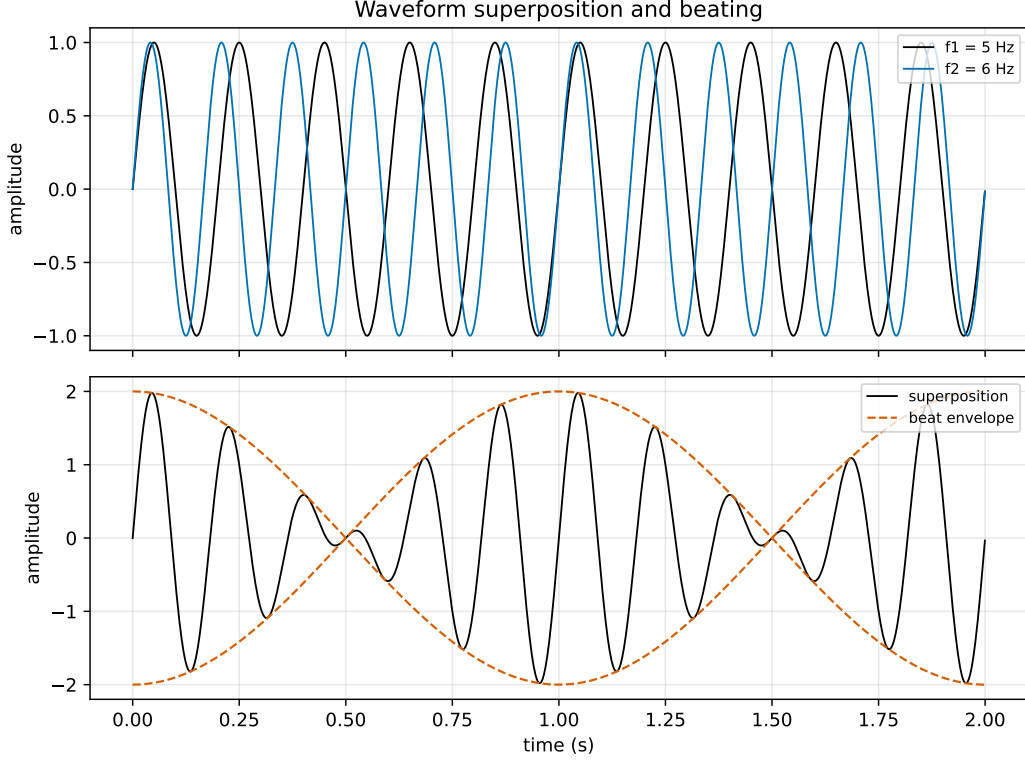


Figure 9: Two sinusoidal components and their time-domain superposition. Nearby frequencies produce a slow beat envelope; equal-frequency addition reduces to the exact phasor law of Lemma 1.

7 Related Work

The lattice model sits in the lineage of the Kuramoto model of coupled phase oscillators [1], whose onset of synchronization and mean-field theory are surveyed by Strogatz [2] and, comprehensively, by Acebrón and colleagues [3]. Our departure is that couplings are signed by a geometric orientation class and the polar class is antiferromagnetic, so the lattice is frustrated: not all pairwise alignment preferences can be satisfied at once. Frustration in this precise sense goes back to Villain’s analysis of competing interactions and spin glasses [7], and it is the mechanism that keeps the phase field in the partially ordered regime of Section 5 rather than at global lock.

The SCM is a coupled dynamical network on a ring and is close in spirit to Kaneko’s coupled map lattices [4], which showed that simple local maps with diffusive coupling generate rich spatiotemporal behavior. The term *stable chaos* is due to Politi and Torcini [5], who reserve it for irregular dynamics in linearly stable systems; we adopt it operationally, as stated in Section 1, without claiming to have measured Lyapunov exponents. Finally, the SCM’s two-channel, opposition-plus-noise design was motivated in part as a response to probabilistic-bit (p-bit) computing [6], which builds computation from stochastic binary units; the SCM instead keeps a bounded continuous state and studies the persistence and occupancy that antagonistic and imitative coupling produce together.

8 Discussion and Future Work

The two models support a single, deliberately modest claim: antagonistic and imitative coupling, each impoverished on its own, together sustain bounded, non-converging dynamics that remain both persistently mobile and cross-node diverse. As the ablations show, occupancy entropy alone does not certify this, since an unanchored herd also visits much of the space; the operational signature is instead the conjunction of liveness and diversity. We have stated this operationally and backed it with reproducible, seed-controlled figures and metrics; the SCM ablations isolate the two mechanisms, and the lattice sweep isolates frustration. We emphasize the scope. This is a computational exploration of finite discrete and semi-discrete models. We do not take a continuum limit, we do not estimate Lyapunov spectra, and we make no hardware claims; the “chaos” in our title is the operational notion of Section 1, aligned in spirit with but not a measurement of the stable chaos of [5].

Several extensions follow directly from the constructions. The orientation classifier and the lattice already support the toroidal option and dimensions one through three, so the frustrated phase dynamics can be studied on higher-dimensional tori where every node has the full 6/12/8 neighborhood and boundary effects vanish. The exact phasor algebra of Section 6 invites an audio reading of the lattice, mapping node phases to partials and studying the timbral consequences of frustration, and the same orientation-classed couplings suggest a constraint-driven, wave-function-collapse style of spatial generation. Quantifying the SCM’s persistence spectrally through the autocorrelation of Section 2.4, and mapping the full (N, δ) parameter surface, are the natural next measurements. Each of these is a concrete, testable continuation of the same thesis: opposition without imitation freezes, imitation without opposition collapses, and the two together stay alive.

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